Fixed-target excess: initial analytic entry

Research note for independent review

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1 Entry from fixed-target excess

This note isolates the first acquisition step. It does not assert a complete local exclusion or a sharp Berry–Esseen theorem. Let P be a centered, unit-variance law, S_j a sum of j independent copies, and set

$$\rho = \mathbb{E}|X|^3, \quad \beta = \mathbb{E}[X|X|], \quad \gamma = \mathbb{E}X^3, \quad c = C_{\mathrm{E}} = \frac{3+\sqrt{10}}{6\sqrt{2\pi}}.$$

The threshold H always has the units of the raw sum. A fixed-target maximizer maximizes

$$a(P,H) = \sqrt{n}\{\mathbb{P}(S_n \le H) - \Phi(H/\sqrt{n})\} - c\rho.$$

We use the attainment and contact theorem in the companion fixed-target note. In particular its closed-support contact equality, not ratio optimality, is the variational premise here. We consider a maximizer with $a \ge 0$. The actual normalized score is $c + a/\rho$; no upper bound on it is assumed.

Theorem 1 (Initial moment and support entry)**.** *Assume $n \ge 42$, $C_{n-1} \le c$, and the universal structural inequality*

$$\sqrt{n}\,\Delta_n(P) \le \frac{3031}{10000}\left(\rho + \frac{323}{500}\right) \tag{1}$$

for standardized laws with finite third moment. Then a fixed-target maximizer with $a \ge 0$ satisfies

$$1 \le \rho < \frac{7361}{4000} < \frac{1841}{1000}, \qquad |\beta| < \frac{321}{500}, \qquad \operatorname{supp} P \subset (-96/25, 96/25). \tag{2}$$

The theorem is uniform in all real H and in both signs of a support point. It requires neither C_{n-2} nor an excess bound of order $1/n$.

Inequality (1) is an external structural input, recorded with its Shevtsova attribution in the manuscript introduction. It is not proved by the exact arithmetic below. Only this one of the two affine structural bounds is needed for the displayed moment radius.

Proof. Since $c\rho \le \sqrt{n}\Delta_n(P)$ and $c > 819/2000$, elementary division gives

$$\rho < \frac{(3031/10000)(323/500)}{819/2000 - 3031/10000} = \frac{7361}{4000}.$$

The other affine bound $0.3322(\rho + 0.429)$ alone would give the larger number $712569/386500$; it must not be used to claim $7361/4000$. Put $m = \mathbb{E}X_+ = \mathbb{E}X_- > 0$ and $a_\pm = \mathbb{E}X_\pm^2 = (1 \pm \beta)/2 > 0$. Cauchy on each sign class gives $m\rho \ge a_+^2 + a_-^2$ and $m^2 \le a_+ a_-$. Hence

$$\rho \ge \frac{1+\beta^2}{\sqrt{1-\beta^2}}.$$

This function increases with $|\beta|$. For $r = 1841/1000$, $b = 321/500$,

$$(1+b^2)^2 - r^2(1-b^2) = \frac{93288849}{50000000000} > 0,$$

which proves the second assertion without assuming bounded support.

Write $v = n-1$, $s = \sqrt{n/v}$, $F_v(x) = \Phi(x/\sqrt{v})$ and $e_n = n(s-1)-1/2$. Closed-support contact, the upper predecessor bound $F_{n-1,P}(H-t) \le F_v(H-t) + c\rho/\sqrt{v}$, and $\sqrt{n}(F_{n,P}(H) - \Phi(H/\sqrt{n})) = c\rho + a$ give

$$0 \le H_n(H,t) + c\{-|t|^3 + \tfrac{3}{2}\rho t^2 + 3\beta t + \rho e_n\} - na \quad (t \in \operatorname{supp} P), \tag{3}$$

where exactly

$$\frac{H_n(H,t)}{n^{3/2}} = F_v(H-t) - F_{v+1}(H) + tF'_{v+1}(H) - \frac{t^2-1}{2}F''_{v+1}(H).$$

The extra $-na$ can be discarded in this upper bound. This is the precise reason the predecessor assumption is $C_{n-1} \le c$, not a comparison with the actual score. No continuity assumption on the distribution function at $H-t$ is inserted.

Rationalization gives

$$ne_n = \frac{s^2(2s+1)}{2(s+1)^2}, \qquad s < 81/80 =: S, \qquad ne_n < \frac{793881}{2073680} < \frac{48}{125}.$$

The first expression increases for $s \ge 1$, as its logarithmic derivative is $2/s + 2/(2s+1) - 2/(s+1) > 0$. Thus $e_n < 8/875$.

For completeness we rederive the Gaussian estimate used at the radius. The heat identity $\partial_v F_v = F''_v/2$ permits spatial expansion of $F_v(H-t)$ to order four, variance expansion of F_{v+1} to order two, and variance expansion of F'_{v+1}, F''_{v+1} to order one. The cubic coefficient is $(-t^3+3t)/6$, and the quartic coefficient is $(t^4 - 6t^2 + 3)/24$. Therefore, for $t = \pm u$, $z_v = H/\sqrt{v}$,

$$H_n(H, \pm u)/s^3 = \pm A(1 - z_v^2)\phi(z_v) + B\phi'''(z_v) + \mathcal{E}, \tag{4}$$

where $A = (u^3 - 3u)/6$, $H_4 = u^4 - 6u^2 + 3$ and $B = H_4/(24\sqrt{v})$. The four error bounds after division by s^3 are respectively

$$\frac{u^5\|\phi^{(4)}\|_\infty}{120v}, \quad \frac{\|\phi^{(5)}\|_\infty}{48v^{3/2}}, \quad \frac{u\|\phi^{(4)}\|_\infty}{8v}, \quad \frac{(u^2-1)\|\phi^{(5)}\|_\infty}{16v^{3/2}}.$$

Gaussian Fourier inversion gives norms at most $3\phi(0) < 6/5$ and $8/\pi < 51/20$. Since $v \ge 41$ and $\sqrt{v} > 6$, for $u \ge 5/2$ this gives

$$|\mathcal{E}| < E(u) := u^5/4100 + 3u/820 + 17(3u^2-2)/78720.$$

These errors are uniform in H; they use no higher moment of P.

If $A, B > 0$ and $B \le A/4$, the leading part of (4) is at most $\phi(0)(A + 9B^2/(4A))$ for both signs. Indeed, for the positive sign and $|z_v| \le 1$, it is at most $\phi(0)(A - Az_v^2 + 3B|z_v|)$, and complete the square. Outside that interval the first term is nonpositive. For the negative sign the first term is at most $2e^{-3/2}A\phi(0) < A\phi(0)/2$. In both cases use $\|\phi'''\|_\infty < 2\phi(0)$: on $[0, \sqrt{3}]$ the polynomial $3x - x^3$ is at most two, and beyond $\sqrt{3}$ use $x^3 e^{-x^2/2} \le 3\sqrt{3}e^{-3/2} < 2$.

For frozen r, b above define

$$Q(u) = -u^3 + \tfrac{3}{2}ru^2 + 3bu + (8/875)r,$$
$$P(u) = \tfrac{2}{5}S^3\left(A^2 + \frac{H_4^2}{256 \cdot 41}\right) + S^3E(u)A + \frac{819}{2000}Q(u)A.$$

At $R = 96/25 < 6$ we have $A > 0$, $0 < H_4 < 36A$, hence $B < A/4$. The sign $Q(R) < 0$ is checked before replacing c by its lower endpoint. The upper bound in (3), multiplied by A, is then strictly below $P(R)$, and exact arithmetic gives

$$-P(R) = \frac{2214258283536204102723}{10250000000000000000000} > \frac{1}{5}.$$

To cover each entire exterior ray, differentiation of the exact Gaussian expression gives

$$|\partial_t H_n(H,t)| < \tfrac{2}{5}(t^2/2 + 1/2 + e_n).$$

For verification put $y = (H-t)/\sqrt{n}$, $z = H/\sqrt{n}$; the derivative is $n\phi(z) + Ht\phi(z) - ns\phi(sy)$. The second-order Taylor error between $\phi(y)$ and $\phi(z)$ costs $\phi(0)t^2/2$, while $\partial_s[s\phi(sy)] = -\phi''(sy)$ costs $\phi(0)n(s-1)$. Define $L(u) = \frac{2}{5}(u^2/2 + 1/2 + 8/875) + (819/2000)Q'(u)$. At R, the six numbers Q, Q', Q'', L, L', L'' are strictly negative. Since $Q''' = -6$ and $L'' = 2/5 - 6(819/2000) < 0$, all their required signs persist for $u \geq R$. Thus both radial upper bounds decrease from a strictly negative boundary value. The contact inequality excludes every support point with $|t| \geq R$. □

2 Moments and an actual high-frequency event

Higher moments are used only after Theorem 1. For $x = |X|$ the nonnegative measure $d\mu_0 = (x-1)^2(x+1/2)\,dP$ has mass $\rho - 1$. Direct expansion gives

$$M_4 - 1 = \int \frac{(x+1)^2}{x+1/2}\,d\mu_0,$$
$$M_5 - 1 = \int \frac{x^3 + 2x^2 + 3x + 3/2}{x+1/2}\,d\mu_0,$$

where $M_j = \mathbb{E}|X|^j$. Both ratios increase for $x \geq 0$: they equal $x + 3/2 + 1/[4(x+1/2)]$ and $x^2+3x/2+9/4+3/[8(x+1/2)]$, whose derivatives are nonnegative. Writing their values at $R = 96/25$ as $k_4(R), k_5(R)$, we obtain

$$M_4 \leq \min(R\rho, 1 + k_4(R)(\rho-1)), \quad M_5 \leq \min(RM_4, 1 + k_5(R)(\rho-1)), \quad M_6 \leq RM_5. \tag{5}$$

Also $\gamma^2 \leq M_4 - 1$, by expanding $\mathbb{E}(X^2 - 1 - \gamma X)^2 \geq 0$.

There is no need to assume a central support gap before the first Fourier bound. For independent copies let $E = \{(X - X')^2 \geq 9/25\}$ and let M be any valid fourth-moment ceiling. Then

$$\mathbb{P}(E) \geq \alpha := \frac{(2 - 9/25)^2}{2M + 6} > 0, \quad |X - X'| \geq 3/5 \text{ on } E, \quad \mathbb{E}[(X-X')^2 \mid E] \leq 2/\alpha. \tag{6}$$

Indeed $\mathbb{E}[(X-X')^2; E] \geq 2 - 9/25$ and $\mathbb{E}(X-X')^4 = 2M_4 + 6 \leq 2M + 6$, so Cauchy proves the mass bound. The conditional second-moment bound follows from its unconditional value two. This is an event under one actual iid marginal, not a replacement characteristic function.

3 A sharper first license

The retained fixed-target conserved identity is

$$(n + H^2)a + nc\rho - \frac{n\phi(H/\sqrt{n})}{6}(\gamma + 3J) + n^{3/2}\mathcal{L} = \mathcal{R}_c(H). \tag{7}$$

Put $\tau(t) = \mathbb{E}[X\mathbf{1}_{X\ge t}]$ and $\nu = \mathcal{L}(H - S_{n-1})$. Over complementary support gaps, define

$$J = \sum \tau_{u,v}(v-u)^2, \qquad \mathcal{L} = \sum \tau_{u,v} \int_{[u,v)} (t-u)\, d\nu(t) \ge 0.$$

The half-open interval includes the correct endpoint atoms. The elementary gap inequality gives $D = (3+\sqrt{10})\rho - \gamma - 3J \ge 0$. In particular the right side of (7) is nonnegative when $a \ge 0$, since it can also be written as

$$(n+H^2)a + nc\rho(1-e^{-w}) + n\phi(0)e^{-w}D/6 + n^{3/2}\mathcal{L}, \qquad w = H^2/(2n).$$

Lemma 2 (Moment-cell license)**.** *Under (7), with $D, \mathcal{L} \ge 0$, suppose $n \ge N$, $\rho \in [l,u]$, $a \ge 0$, and $0 < c_- \le c$. If a bound valid for every real H gives*

$$\mathcal{R}_c(H) \le T < Nc_-l, \tag{8}$$

then either $T < 0$ and this cell is excluded, or

$$\gamma + 3J > 0, \qquad e^{-H^2/(2n)} \ge 1 - \frac{T}{Nc_-l} > 0, \tag{9}$$

$$(n+H^2)a + \frac{n\phi(0)}{6}D + n^{3/2}\mathcal{L} \le \mathcal{R}_c(H) \le T. \tag{10}$$

Thus the first gate need not be the stronger uniform condition $T < 2N/5$.

Proof. For a nonexcluded cell $T \ge 0$. Divide (7) by n:

$$\frac{\phi(H/\sqrt{n})}{6}(\gamma + 3J) = (1 + H^2/n)a + c\rho + \sqrt{n}\mathcal{L} - \mathcal{R}_c/n \ge c\rho - T/n > 0.$$

This proves the sign. Since $D \ge 0$, $\gamma + 3J \le 6c\rho/\phi(0)$; hence $c\rho e^{-w} \ge c\rho - T/n$, proving (9). Finally rewrite the conserved expression as the left side of (10), plus $n(1-e^{-w})(c\rho - \phi(0)D/6)$. Its last coefficient is $\phi(0)(\gamma+3J)/6 > 0$, so it can be dropped. □

The positivity argument uses no small-defect premise. The input $D \ge 0$ is used for the density and unweighted-resource conclusions, not for the first sign deduction. Thresholds of either sign and $H = 0$ are included. If only $T \le Nc_-l$ is proved, the strict sign and positive density floor do not follow. The strict gate is a genuine acceptance condition.

Lemma 3 (First phase price without a small defect)**.** *Under Lemma 2, put*

$$\mathcal{U} = (n+H^2)a + n\phi(0)D/6 + n^{3/2}\mathcal{L}, \qquad q_{\mathrm{ph}} = \tfrac{1}{2}(c_-l - T/N) > 0.$$

Then $\mathcal{U} \le \mathcal{R}_c(H) - q_{\mathrm{ph}}H^2$. In particular a complete polynomial bound $\mathcal{R}_c(H) \le A + F|H| - qH^2$ with $F \ge 0, q > 0$ gives

$$\mathcal{U} \le A + \frac{F^2}{4(q+q_{\mathrm{ph}})}.$$

This improved bound is on $\mathcal{U}$, not on the raw $\mathcal{R}_c$.

Proof. Let $M_c = c\rho - \phi(0)D/6$ and $w = H^2/(2n)$. The exact identity and $\mathcal{R}_c \le T$ imply $e^{-w}M_c \ge c\rho - T/n > 0$. Therefore

$$n(1-e^{-w})M_c = n(e^w - 1)e^{-w}M_c \ge \frac{H^2}{2}(c\rho - T/n) \ge q_{\mathrm{ph}}H^2.$$

Here $e^w - 1 \ge w$ for $w \ge 0$. Subtract this term from the exact decomposition $\mathcal{R}_c = \mathcal{U} + n(1-e^{-w})M_c$, then complete the square. There is no premise $D < 2$, and no multiplication of separate lower estimates on M_c and e^{-w}. □

Lemma 4 (Normalized first-cover consequence)**.** *Under the preceding hypotheses, suppose a separate complete enclosure proves $\mathcal{R}_c(H)/\rho < 10$ for all real H. For $n \geq 42$,*

$$e^{-H^2/(2n)} > \frac{7199}{17199} > \frac{209}{500}, \qquad \frac{D}{\rho} < \frac{25000}{6979}, \qquad \frac{a}{\rho} < \frac{10}{n}.$$

In particular the first license is discharged. The actual score satisfies $c + a/\rho < c + 10/n$, but this is an output, not an entry premise.

Proof. Use the pointwise version of the preceding proof with $T = 10\rho$: $10 < 42(819/2000) < nc$. The density floor is strictly larger than $1 - 10/(42c) > 1 - 10/[42(819/2000)] = 7199/17199$. From (10) and $\phi(0) > 997/2500$, $D/\rho < 60/(42\phi(0)) < 25000/6979$. The nonnegative term $(n + H^2)a$ gives the final bound. □

4 The first scalar enclosure and its remaining gate

This section states exactly what a bounded arithmetic cover must supply. It proves the algebraic reduction, rather than treating an archived PASS as an analytic premise. Fix a closed cell $[l, u] \subset [1, 7361/4000]$. Choose outward ceilings M, g, E_5, b for $M_4, |\gamma|, M_5, |\beta|$ using (5), $|\gamma| \leq \rho$, Pearson, and the sign-moment inequality. In particular $M < 11.1$, $g < 1.85$ and $M_6 \leq RE_5$.

Write $J_1 = \sum \tau_{a,b} \int_a^b (t-a)t\,dt$ and $J_2 = \sum \tau_{a,b} \int_a^b (t-a)t|t|\,dt$. Set

$$K_1 = \mathbb{E}[X|X|^3] - 3\beta - 3\rho\gamma/2,$$
$$K_q = M_5/4 - \rho M_4/2 - 3\beta\gamma/2 + \rho/2, \qquad U = J_1 + (M_4 - 3)/6.$$

The exact raw remainder in (7) is

$$\mathcal{R}_c = c(3\beta J/2 + 3\rho J_1 - 3J_2 - K_q) + H(cK_1 + \phi U) - H^2(c\rho + \gamma\phi/2) + E_n.$$

The fixed target c is the coefficient in every term. Its enclosing interval $[819/2000, 20487/50000]$ is not a bound on the actual score.

The elementary gap integrals give $-M_4/6 \leq J_1 \leq M_4/2$, $-3J_2 - M_5/4 \leq 0$ and $J \leq \rho + 1$. The first two follow by integrating, gap by gap,

$$\int_a^b (t-a)t\,dt + \tfrac{1}{2}\int_a^b t^2 dt = (b-a)b^2/2,$$

and its complementary upper inequality; for the negative part of J_2 use $\int_u^v (v-x)x^2 dx \leq \frac{1}{3}\int_u^v x^3 dx$. Consequently a valid constant bound is

$$A_2 = \tfrac{3}{2}b(2u+1) + 2uM - l/2.$$

One may take the smaller of A_2 and

$$A_1 = \tfrac{3}{2}b(2u+1) + \tfrac{3}{2}\Gamma(R,l)(2u) + \tfrac{3}{8}u^2(u+1) + \tfrac{8}{27}u^3 - l/2,$$

where $\Gamma(R, l)$ bounds the nonnegative part of

$$-v^2/4 + (R+l)v/3 - Rl/2 + l^3(2R-l)/(12v^2), \qquad l \leq v \leq R.$$

This is $v^{-2}\int_l^v (R-x)x(x-l)\,dx$. For each realized negative magnitude v, Tonelli bounds all its gap contributions by this one integral; multiplying by v and averaging charges $\Gamma\mathbb{E}X_-^3$. Thus on

negative gaps it charges $3(\rho J_1 - J_2)$ by $3\Gamma(\rho-\gamma)/2$; on positive parts use $t(\rho-t) \le \rho^2/4$, multiply by $(t-a) \ge 0$, and integrate $\sum \tau_{a,b} \int_a^b (t-a)\,dt = J/2$. This gives $3\rho^2 J/8$ and includes the positive part of a central gap with $a < 0$. The remaining term uses $-K_q \le 8\rho^3/27 + (3b/2 - 1/2)\rho$, obtained by $M_5 \ge M_4^{3/2}$ and maximizing $\rho y^2/2 - y^3/4$. Thus $A_* := \min(A_1, A_2)$ is valid when the chosen Γ is certified.

Valid bounds on $|K_1|$ are

$$\begin{aligned} k_{\mathrm{rad}} &= 3ub + \frac{R(R+2-3l/2)}{R+1/2}(u-1), \\ k_{\mathrm{old}} &= 6u - 3u^2/2 + 3b, \\ k_R &= Ru/2 + \max_{t=l,u} |R/2 - 3t/2| g + 3b. \end{aligned}$$

For the first, use the radial identity for $K_1 + 3\rho\beta$ with ratio $x(x+2-3\rho/2)/(x+1/2)$, convex in x and decreasing in ρ. Its negative part is at most $31/40$, so require its stated endpoint ceiling to be at least $31/40$. The second uses $R \le 6$ and $|\mathbb{E}[X|X|^3] - 3\gamma| \le 3\rho$. The third bounds the two fourth semimoments by $R(\rho \pm \gamma)/2$. Take k to be their minimum. Also $U \in [-1/2, 2M/3 - 1/2]$, so use $v_H = (2/5)\max(1/2, 2M/3 - 1/2)$ (or a separately proved smaller bound). Then $q_* := c_- l - g/5 > 0$ is a valid quadratic curvature.

It remains to have a *complete* Fourier bound

$$|E_n| \le K_F + s_F |H|, \quad K_F = \tfrac{16}{25}(L_0 + K_H + L_2/\sqrt{N}), \quad s_F = \tfrac{16}{25} L_H. \tag{11}$$

The compensated quadratic-hinge inversion has factor $2\sqrt{n}/\pi$. The retained triangle transfer uses the parameters N, M, g, M_6 and sixth-order Taylor estimates on $[0, 1/2]$. Its conclusion is

$$I_0 \le (L_0 + L_H |H|)/\sqrt{n} + L_2/n.$$

The retained complete six-part high-frequency lemma uses the actual event (6) and gives $\sqrt{n} I_\infty \le K_H$. Every damping, floor and window gate in that lemma must hold; one takes a minimum only of complete six-term bounds. The event proof does not permit dropping its Gaussian comparator tail. Neither Fourier lemma uses a score bound or a defect feedback hypothesis.

Completion of the square now proves the all-threshold bound

$$\mathcal{R}_c(H) \le T := K_F + c_+ A_* + \frac{(c_+ k + v_H + s_F)^2}{4 q_*}, \tag{12}$$

provided $A_* \ge 0$, $k \ge 0$, $q_* > 0$ and $c \in [c_-, c_+]$. This is a scalar inequality, so using a formal coefficient interval containing c is sound independently of the actual score. The first numerical obligation is a finite closed cover of $[1, 7361/4000]$ on which the stated Fourier gates and $T < N c_- l$ all hold. The endpoint $\rho = 1$ is included; at that point the law is a symmetric two-point law, and all moment ceilings remain well-defined. Adjacent cell endpoints must both be covered.

Scope of this note. The bounded companion checker verifies the support rational reductions, five exact finite-law moment tests, and a negative control rejecting radius three with the same envelope. It does not evaluate a whole raw cover or prove any local comparison predicate. A separate 24-cell calculation reports $\mathcal{R}_c/\rho < 10$; its source, outward arithmetic, coverage and complete Fourier premises require their own review before composition with Lemma 4. Subsequent support contractions, central-gap acquisition, confinement and LOW/HIGH exclusion still require their own ordered hypotheses. In particular the present entry theorem alone does not remove the independent finite base.